3D Bharat

3D Bharat has Four main operational area

1. Desktop application:

2. Central web Panel:

3. Construction Site Tablet

4. Mobile applications

1. Desktop application:
    A. For the Design
    B. For the Measurement

2. Central Web Panel
    A. Platform Owner
        I. Register Departments and Contractors.

        II. Define the State, District, and Taluka hierarchy.

        III. Define templates for various project objects (e.g., bridge footings, roads, culverts, buildings, etc.).
        IV. Manage subscription plans applicable to different stakeholders and user categories.
    B. Department Web
        a. Defined the Employee's
        b. Define Roles & Responsibilities
        c. Register Project
        d. Define Work packets under the project.
        e. Assign project to the contractor for
            I. Design (need the initial 3D Files + total solution, points Data)
            II. Deployment (based on the design, update the on-ground status of the project work)
            III. Construction, Monitoring & measurement's (Bridge, Road, Tunnel, Asset Fixing, Painting) in and / or Mode.
                a) Construction: monitor the daily work progress report form the ground staff.
                b) Monitoring: Periodic drone flying & Capture the 3D files
                c) Measurement: map the Recent Drone 3D files with Previous flying data & Design data. cross verify the measurements with the on-ground data.
            IV. Maintenance

        f. Define Templates: for the ideal work steps & details in the construction, it's separates for bridge, Road, building, tunnel & other operational area's for constructing of the objects like bridge footing, bridge span, Road, underpass etc.)

g. Define the contractor wise work targets
    I. For Allocated work completion time line
    II. Like of equipment's
    III. Safety measures

C. Contractor Web: Contractor can define their own projects, or he can be assigning as a contractor in the department project.

If Contactor creates the project then he can

a. Defined the Employee's
b. Define Roles & Responsibilities
c. Register Project for (Road and / or bridge and / or tunnel, Rail and or bridge and / or tunnel, Metro and / or bridge and / or tunnel)
d. Assign project for
    IV. Design (need the initial 3D Files + total solution, points Data)
    V. Deployment (based on the design, update the on-ground status of the project work)
    VI. Construction, Monitoring & measurement's (Bridge, Road, Tunnel, Asset Fixing, Painting) in and / or Mode.
        d) Construction: Collect the daily work progress report
        e) Monitoring: Periodic drone flying & Capture the 3D files
        f) Measurement: map the Recent Drone 3D files with Previous flying data & Design data. cross verify the measurements with the on-ground data.
    VII. Maintenance

e. Define Templates: for the ideal work steps & details in the construction, it's separates for bridge, Road, building, tunnel & other operational area's for constructing of the objects like bridge footing, bridge span, Road, underpass etc.)
   Define Equipment's

Contractor can also manage subcontractors even work is allocated by the Department.

f. Introduce Sub contractors
g. Define the Work packets & allocate to the sub-contractor.
h. Define / inherit Template's & Link with the objects of the (Road, Bridge, Rail, Tunnel, Metro)

Sub-Contractor can also manage their employees. for the Assigning them targets via web panel, Tablet based work progress monitoring, Mobile app for work activity monitoring.

D. Administration

Kindly keep in mind department District collector, SDM of Taluka are the static post's, platform owner or department can introduce / change the Name & mobile no of the District Collector.

Collector can login in the Web & view the all the projects who are passing from his jurisdiction area.

Project Configuration: In this process there are sub process / modules :

1. On web panel Project owner (Department / Contractor) has to insert the details of the KM & Chainage District Taluka (if available) status of the project (under construction / New / Completed) about the new project.

2. The authorized person has to upload the

   I. KM wise 3D point cloud files (old 3d point cloud files or 3d point cloud files will be captured from the drone) with the GPS & ground point data via web portal.
   II. Will also upload the Drone Route map file, ideal drone flying conditions.

3. Allocate the km wise design work to the designer (user)

4. Designer will download the project, KM 3d point file & other ground point data on the desktop application & Design the objects.

5. once the design is completed on the desktop application then it will upload to the web server.

6. authorized Web user has to verify the design & if required then he can reject the design.
7. authorized web user confirm the design & mark the 3d point cloud file for the merger. Authorized web user can allocate the multiple objects as a merger & allocate the designer for the merger work.

8. allocate merger (multiple design work on the same point cloud) to the design by the web user.

Project Deployment: complete project or part / parts (K.M.) of the project can be assigned for the deployment stage by the web user. Once the project / part of the project is under the deployment. Then Department can allocate the contractor as a deployment agency. If the project is New then web user can skip the deployment & directly go to the construction.

Define the Template for work execution sequence. Web user (from the project owner (department / contractor) can select the pre-configured template or can define

Deployment contractor will allocate the work to their employee to get the details of on-ground status of the designed objects by defining the construction blocks in case of the road, railways, tunnels earth work. Object's for the bridge, & metro.

Allocated on ground staff will use the tablet application to define the road block's & object status with the current status of the road block & object as per the linked template.

Once the ground staff update the road construction block & status of the objects as per the templates, status of the assets as per the design then Web user (authorized from the Deployment contractor and / or Project owner) will cross verify the updates & once he confirms then system accepts the deployments.

Kindly note KM wise (including Road, Bridge, under pass etc.) deployments will be confirmed by the web user is must.

If No work is done in the particular KM then web user can directly change the status of the KM to the construction.

If the particular KM work is completed including the assets are fixed then web user can change the status of the KM to the maintenance.

Once the KM deployment is confirmed then particular KM or group of the KM can be allocated to the drone contractor. Kindly note drone contractor cannot be introduced in the system if the particular KM drone travel path & the Drone flying conditions is not uploaded by the Web user (deployment contractor / project owner)

Drone contractor will be defined with the KM or group of KM's, drone flying with the time of the periodic, upload lead time has to be configured.

One drone contractor can have multiple group of KM work in one project. With different settings.

When the status of the Project Km / Group of KM 's (may be consider as a Packet) is "Construction" then system can be introducing the construction contractor, asset (electric fitting, painter, any kind of signage, etc.) contractor.

Against the construction project owner (department / contractor) can introduce the measurement contractor. This measurement contractor staff will do the measurements on the point cloud files uploaded by the drone contractor. Measurement contractor web user will review the measurement done by the desktop engineers. once web user approves this then it will be visible to the concern web user of the project owner (department/ contractor) with the respective quantities.

Project Owner (department / Contractor) will be able to view the periodic work progress by overlapping the different drone KM image measurement. Also, system will be able to show the KM wise work completion status. If in future project owner (department / Contractor) enters the targets against KM wise road, objects, assets then we can map our data & able tell them the project is running on the time or delayed of ahead of the time frame.